

# Hybrid Deep Reinforcement Learning Model for Safe and Efficient Autonomous Vehicles at Pedestrian Crossings

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**Abstract**—The autonomous vehicle is a particularly promising area of application for machine learning algorithms. Autonomous driving is challenging due to the complexity of the road network, the hardly predictable human behaviors, either pedestrians or drivers and the imperative need to ensure the safety of all road users. Deep Reinforcement Learning algorithms have garnered significant interest as a viable option for controlling autonomous vehicles. They allow defining a control policy by solely describing the observation space, the action space, and a reward function. Yet, the most common solutions face difficulties when the action space is hybrid (simultaneously continuous and discrete), and the reward is sparse, which is the case when a selection between choices must be made (like yielding or not), and the trajectory of the vehicle must be adapted according to this decision. This paper addresses these challenges by focusing on controlling an autonomous vehicle at pedestrian crossings. The vehicle adapts its speed (continuous decision) and the displayed signal (discrete decision) according to the pedestrian's state. The unique characteristic of the speed profile is to facilitate safe pedestrian crossings without requiring the vehicle to come to a complete stop. This raises the consistency problem between the speed profile and the signals issued by the vehicle. This paper introduces a new hybrid DRL model capable of making both decisions. It discusses the training method, the construction and necessary constraints on the reward, as well as the success of generalization to different speed limitations.

**Index Terms**—Autonomous vehicle, Hybrid Deep Reinforcement Learning Model, Hybrid systems, Pedestrian crossing.

## I. INTRODUCTION

The autonomous vehicle (AV) domain has seen immense technological advances recently. The contribution of machine learning techniques to this domain is incontestable, improving the perception and enhancing the AV decision-making and control. We are getting closer to seeing AV on

the road or industrial sites. Yet, some situations remain challenging for AVs, especially when vulnerable road users, like pedestrians, are present. When a pedestrian interacts with an AV (e.g. at a pedestrian crossing), the vehicle must be able to ensure the safety of the pedestrian according to its capabilities. Many studies focus on predicting the trajectory of pedestrians by taking into account worst-case scenarios, estimating the uncertainty of their trajectory, or taking into account inter-agent communication [10].

The means of communication between the AV and the pedestrian are multiple. The literature mainly focuses on a luminous signal that announces the vehicle's decision, as discussed in the work of Rasouli et al. [15]. For instance, the signal displayed on the front of the vehicle typically consists of a red light, indicating that the vehicle has the right of way, and a green light, indicating that the vehicle must yield. This kind of signal is displayed on the vehicle (windshield, headlights, bonnet) or projected onto the road [14]. Some other works exploit the speed profile of the vehicle, with an incentive deceleration, to let the pedestrians know that they can cross safely [21]. Although vehicle speed profile is widely known as a determinant of pedestrian decision [1], [9], [22], to the best of our knowledge the first work that exploited this property is presented in [6]. The authors propose a rule-based system to improve vehicle-pedestrian interaction. A horn was needed to avoid collisions. The approach was enhanced using optimal trajectory control of two conflicting agents [19]. Variations in characteristics between pedestrians prompted the introduction of Deep Reinforcement Learning (DRL) for speed control in [20].

By merging these different means of communication (the signal and the incentive speed profile), it can be expected to secure the interaction between the AV and the pedestrian while lowering the crossing delays and reducing energy

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consumption [5]. However, it is challenging to design a policy able to handle this kind of interaction with the pedestrian, as it is composed of a discrete action (the signal) and a continuous action (the speed profile), both having an influence on the behavior of the pedestrian and thus an impact on the efficiency and the safety of the interaction. In this paper, we propose an architecture based on hybrid DRL models designed for the context of pedestrian-vehicle encounters. The vehicle must react according to the danger of a pedestrian crossing: decision-making, speed management, accident prevention, and coherent communication.

This paper is organized as follows. The section II describes the studied use-case and presents a brief overview of the usable techniques to manage the resulting situation. It follows section III, which details the proposed innovative approach based on hybrid DRL. Section IV describes the simulation environment and gives the results of the proposed approach. finally, the results are discussed in section V.

## II. PEDESTRIAN CROSSING AND AUTONOMOUS VEHICLES

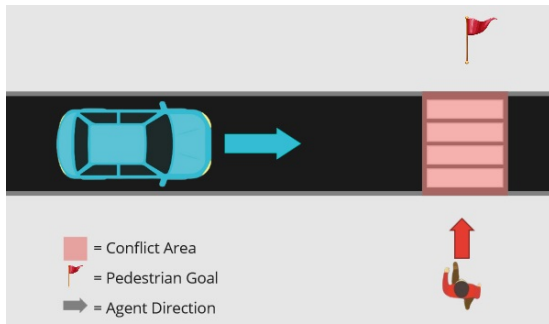


Fig. 1: Environment scheme.

The studied scenario is a single CAV-pedestrian encounter at a crosswalk on a one-way urban road. The initial state of the environment is defined at the moment when the car detects the pedestrian. We assume the pedestrian is always willing to cross. In this scenario, the conflict between the AV and the pedestrian paths may appear, and both road users must cooperate to safely and efficiently pass the crosswalk.

The pedestrian is considered as an agent whose personal objective is to cross the road, avoiding being in collision with the AV. To decide whether he/she can cross, the pedestrian analyzes the environment and chooses according to the gap between his crossing time and the vehicle's arrival time [12]. We also assume that the pedestrian is always walking perpendicularly to the road.

AV speed management can rely on many techniques. The simplest one is to trigger the safety system that abruptly brings the vehicle to a complete stop before the crosswalk. However, more communicative and reassuring approaches are needed. To this end, the AV can be controlled by exact solutions such as Quadratic Programming. Unfortunately,

those solutions are not highly responsive to the randomness of pedestrian behavior, making them hard to use in real-time scenarios. The DRL methods are more responsive and performant, using continuous action space-based policies such as DDPG and PPO [5], [11], [17], [21]. In this paper, we consider that it is possible to separate the AV decision-making on the arrival sequence of agents (i.e. should the vehicle yield or not) from its speed control. This involves a hierarchy between the discrete decision and the continuous control. Many DRL methods can be applied with the above criteria, named thereafter as hybrid methods. The obvious methods are the discretization of continuous variables or the relaxation of discrete variables. This is a way to deviate from the hybridization problem to make the model work. The model is thus no longer a hybrid model but uses techniques/methods on spaces of continuous or discrete actions.

Hybrid models, by definition, consider both the continuous and discrete action space. An overview of these models of the literature are presented below.

The **Multi-Task Learning** consists in managing several tasks at the same time. The main example for this idea is the Hindsight Experience Replay (HER) method [2]. HER separates the different objectives of the reward to promote targeted actions, despite the episode's failure in its realization. It converges more quickly toward the complex actions we want to achieve. To do this, it first separates the reward into several distinct objectives (multi-task). Then it stores the rewards of the different tasks in a replay buffer (separately). HER is however not very efficient against shaped rewards (discrete rewards). Designing a non-sparse/shaped reward related to the model's decision-making in our environment is very difficult. Moreover, to define the importance of each part of the reward, the only solution is to modify the frequency of their appearance in the replay buffer.

Another solution called **Parameterized Action Space** consists in hierarchically classifying discrete and continuous actions in the same model: a discrete action is linked to a set of continuous actions. The two following models start from DRL models on discrete or continuous actions only to build a parametric model on top. The action-space hierarchy expresses the dependency between discrete and continuous actions. In counterpart, the approach significantly increases the number of outputs of the model.

The **parameterized DDPG** [7] transforms the actor of the DDPG to ask him in the output: both discrete choice and the continuous values relative to the various discrete choices. All these data are then returned to the critic, which, just like the DDPG, returns as output an estimate of the reward with respect to the actions and the given state. It uses an epsilon-greedy degressive exploration on discrete actions and continuous actions with uniform laws. The P-DDPG model thus proposes an actor model able to perform a discrete choice and to optimize the continuous variables related to this choice.

The **parameterized DQN** [18] uses the same structure

as the DQN to determine the discrete action to be carried out with the help of an estimate of the reward which one seeks to optimize ( $Q(a,s)$ ). However, it adds an actor that calculates the continuous variables in the same way as a classical DDPG. In this case, the only difference is that it relies on the critic (model estimating  $Q$ ) to achieve a good classification. It eventually compares the rewards obtained according to the discrete choice. The article also describes the techniques for limiting the values of the gradients. The squashing technique is the one that gives the best results in the tests we have been able to perform. This supports the theory of the article proposing the PATRPO model [8], which states that squashing is more interesting in learning than invert-gradient. This article also introduces multiple models, such as parameterized TRPO and SVG. Unfortunately, it does not explain how the TRPO-based algorithm is built [16]. Referring to the construction of the PASVG, it uses a two-step actor, one for the discrete part and another for the continuous part, with the discrete choice as an argument.

Inspired by the literature, in the next section, we present a new hybrid model and its application to the use-case of pedestrian crossing, splitting the discrete decision from the continuous control.

### III. HYBRID DEEP REINFORCEMENT LEARNING FOR PEDESTRIAN CROSSING

#### A. Design of the Hybrid Model

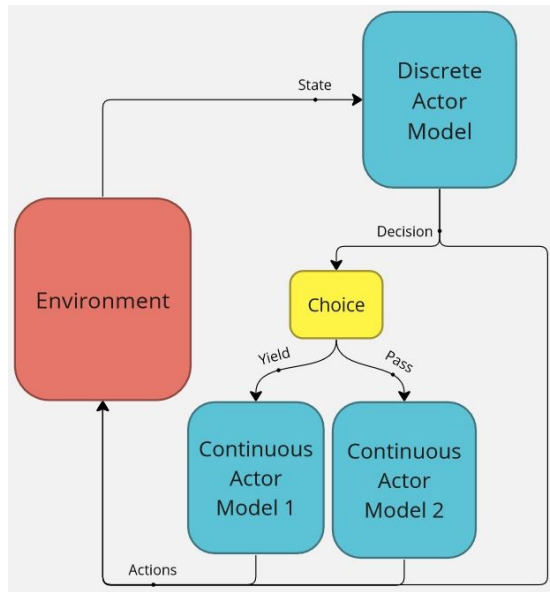


Fig. 2: Diagram of the hybrid model.

We separate a state space between a car going first and a car yielding. The difficulty of managing the trajectory varies enormously. During training, collisions happen much more often when the car has to yield. This extremely important bias in the environment prevents the algorithm

from converging easily. Our first models were focusing on step-by-step decision-taking to be able to quickly change strategies if the pedestrian refuses to cooperate. These models turned out to be very unstable. The two scenarios were trained together with such a solution, but the reward balance between the two scenarios was impossible due to a certain hierarchy of goals in it. The accident occurrence should always be prioritized. It results in a model learning only on the second scenario, i.e. the scenario that impacts the reward the most.

The model presented in this article bypasses the problems encountered above thanks to task prioritization. First, the hybrid action model must make a one-time decision: go first or let the pedestrian cross.

Then it must act accordingly to the plan by controlling the speed correctly. The proposed hybrid model makes the learning of the continuous/discrete parts quite independent. The discrete model chooses which continuous model will have to act on the vehicle controls. This allows the display of the decision of the vehicle via a light signal to communicate with the pedestrian.

We could use a non-supervised classification algorithm for the decision-making, but we decided to use another DRL model as it can be expected to give better results. The PPO is able to perform classification despite the independence of the steps in our learning batch (1 step per episode) as shown in the paper [8]. Indeed, PPO uses policy optimization. The continuous part is also a PPO model for this specific problem, giving good results, as shown in [5].

According to this architecture (Fig. 2), the decision-making part and the speed control of each scenario (yielding or not) need their own reward, described thereafter.

#### B. Reward Definition of the Discrete Model

The goal of the reward of the discrete (i.e. decision-making) part is to ensure that the AV establishes a correct decision regarding the initial situation.

The reward is as simple as possible to express the intrinsic will of the vehicle. Indeed, the objective of this part is to know the capabilities of the AV in order to make an adequate decision. A sparse reward is used, with a penalty if and only if the scenario fails (i.e. the AV is dangerous or his communication is inconsistent). At the beginning of the scenario, we decide which strategy to adopt towards the pedestrian, and at the end of the scenario, we analyze the trajectory to see if the decision was correct. The benefits of such an algorithm are multiple: less computation per step, better convergence/performance, and a specific reward for decision-making.

To evaluate the decision-making model, the focus is put on two main points:

- 1) the AV must be able to realize its decision (a penalty is applied if the continuous model fails to apply the decision)
- 2) the AV must not endanger the pedestrian, even if they were to change their velocity

Thus, the reward  $R_d$  can be written:

$$R_d = -\mathbb{1}_{dangerous} - \mathbb{1}_{inconsistent}$$

where  $\mathbb{1}_{dangerous} = 1$  if the worst-case scenario causes an accident.  $\mathbb{1}_{inconsistent} = 1$  if the CAV and the pedestrian did not enter the crosswalk in the proper sequence.

This model does not require a balance between the two scenarios. Indeed, thanks to an exploration parameter on the decision-making of the discrete model (normal distribution), each continuous model is improving. It is important to recall that each scenario does not converge at the same speed. The exploration parameter must be adjusted accordingly. In our case, the yielding scenario is more complex and will require more steps to converge. Despite the absence of constraint and the unbalanced complexity between the scenarios, the algorithm chooses correctly without favoring a single choice.

If the continuous models are trained, the decision distribution is only dependent on the discrete reward. The reward must then be chosen wisely to prevent a dominant choice.

### C. Reward Definition of the Continuous Model

The goal of the reward of the continuous model is to ensure that the AV applied the correct behavior toward the pedestrian according to the plan. It depends on the decision of the AV. If it is decided that the pedestrian should go first, the actions are penalized according to a safety criterion and according to the speed of the vehicle. If the AV decides to go first, we penalize only the actions which do not respect the speed limit.

The speed reward is the only one used in this case. We use the squared speed to penalize in priority the low speeds and to have an appropriate speed curve with:

- 1) a constant deceleration
- 2) a stable speed allowing the pedestrian to pass
- 3) an acceleration until the speed limit

The reward for safety is calculated to penalize the reward exponentially with respect to the distance to the point of no return, i.e. the position where the car cannot stop in an emergency. The reward has a small/no penalty when the car still has 1s of reaction time before it has to make an emergency stop. A trick was also used to make the safety reward converge more efficiently: the worst safety reward is stored and applied for the rest of the episode, creating a no-return effect (amplified by the RL mechanism). The two rewards are adjusted to help the algorithm converge easily despite large differences in the environment data.

Thus, the reward  $R_c$  for the continuous part can be written:

$$R_c = -40e^{-4(\frac{D}{V_m}-1)} - 20\frac{(V_c - V_m)^2}{V_m^2}$$

where  $D$  is the minimal distance to stop,  $V_m$  is the car speed limit and  $V_c$  is the actual car speed.

## IV. SIMULATION AND RESULTS

### A. Simulation Environment

The simulation is based on a Markov Decision Process (MDP) in order to apply the DRL models. The reward has already been described, but the state and action space must also be defined. The action space includes the decision-making and the AV acceleration. The parameters composing the state space are an adapted version of [5]. It provides information on the pedestrian, the vehicle, and the crosswalk environment states (Fig. 3). All the simulation is implemented in Python with the Pytorch library and the Gym OpenAI toolkit [4].

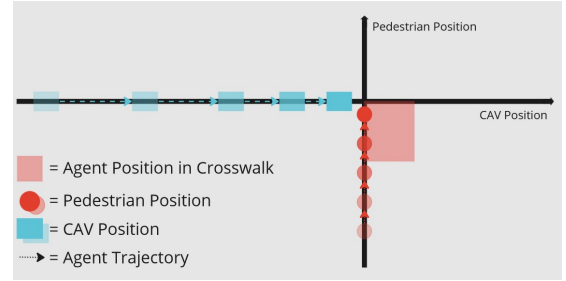


Fig. 3: Simulation of the environment.

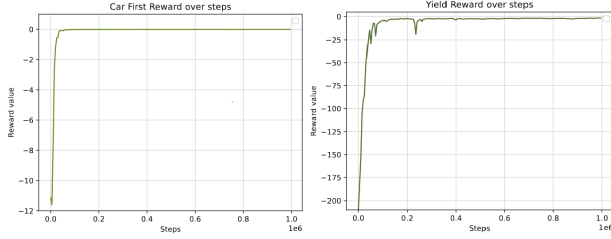
The initialization is also very important to set up. The results of the scenario heavily depend on this setting, as it requires a very large scenario initialization space. The initialization of the scenario uses a continuous uniform distribution of the parameters. The parameter intervals used to define the initial environment are described in Table I. Initially, the vehicle is always driving at the maximum speed allowed on the road with a constant speed. This initial speed varies from  $5m/s$  to  $15m/s$ . The pedestrian speed range is estimated thanks to a literature aggregation about pedestrian speed [3]. The simulation model considers other pedestrian features such as reaction time and safety margin. Both are randomly distributed according to data given by [1], [13]. For safety considerations, two "abnormal" pedestrian behaviors are randomly selected. The first one is the complete stop on the crosswalk, with a probability of 2% for each 0.3s. This intends to simulate falling or any other problem the pedestrian faces during the crossing task. The second behavior is the unauthorized cross (red sign) if the pedestrian estimates that the safety margin is acceptable.

TABLE I: Scenario initialization parameters

| Name                | Initial values | Units   |
|---------------------|----------------|---------|
| Car Acceleration    | [0.0; 0.0]     | $m/s^2$ |
| Car Speed           | [5.0; 15.0]    | $m/s$   |
| Pedestrian Speed    | [0.75; 1.75]   | $m/s$   |
| Pedestrian Position | [-3.5; -0.5]   | $m$     |
| Road Size           | [2.5; 3.5]     | $m$     |



(a) Discrete model.



(b) Continuous model.

### B. Analysis of the Efficiency of Hybrid DRL

The three different models are converging. The discrete model (Fig. 4a) needs at least 600000 time-steps to converge. The larger is the range of the initialization parameters, the longer is the convergence time. The first scenario (Fig. 4b) with the car passing first is converging quickly due to the simplicity of the task requested (keep the same speed). However, the second scenario (Fig. 4b) takes longer. Even when the reward appears to have converged, this scenario continues to improve to 800,000 time-steps.

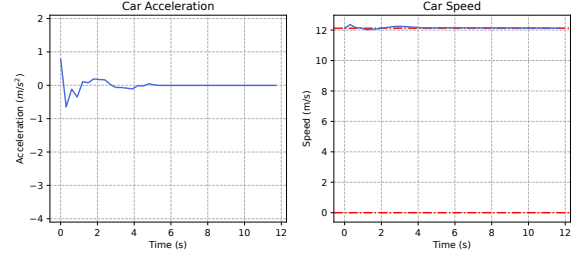
The algorithm is performing very fast even though the range of scenario initialization parameters has been widely opened. This ability to adapt to a wide range of data is made possible by weighting the reward against the speed limit. Otherwise, the algorithm was only learning the scenarios with the highest speed limits.

We observe a shift in the proportion of the scenarios during training. The yielding decision occurs more and more often: from 50% (random choice) to 70% on average.

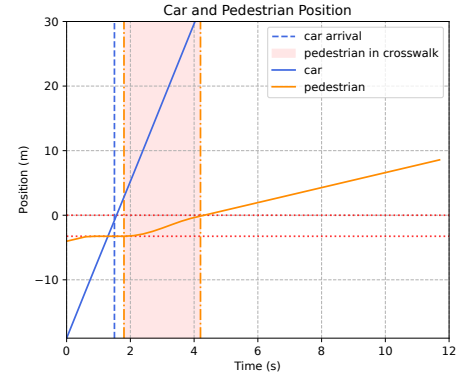
We train and test the model with a pedestrian capable of falling while crossing to validate the model's capabilities. The trained model is very performant, with no occurrence of accidents and a good reactivity to pedestrian actions. The AV is able to respect the safety distance while performing an efficient crossing.

### C. Action Analysis of Hybrid Model

To visualize the actions of the model on a standard episode, we provided an example of each scenario (Fig. 5 and Fig. 6).

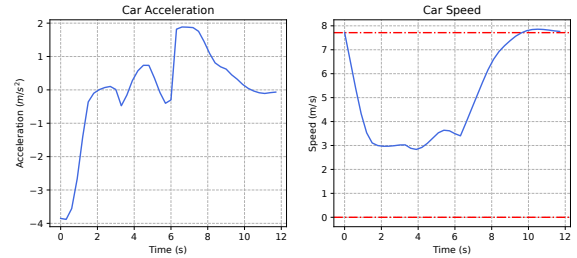


(a) Speed and Acceleration of the car versus time.

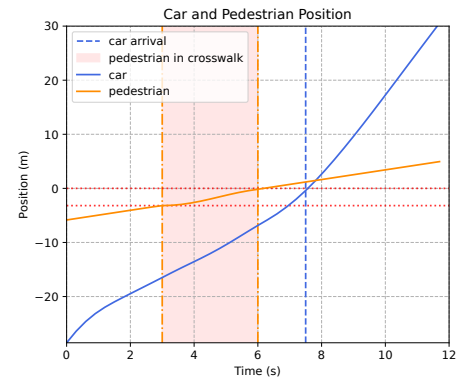


(b) Car and Pedestrian Position versus time.

Fig. 5: Scenario of an AV going first



(a) Speed and Acceleration of the car versus time.



(b) Car and Pedestrian Position versus time.

Fig. 6: Scenario of an AV yielding to the pedestrian

For the scenario where the car is the first to enter the crosswalk, the AV speed and acceleration must be very constant and stable, which is the case in figure 5a. In this example, the pedestrian must wait almost 2s for the AV to pass before entering the crosswalk (Fig. 5b). The discrete model chooses to pass because the AV can't stop before the crosswalk. In this case, the safer decision is to prevent the pedestrian from crossing. If the AV tries to stop, it gives a wrong signal to the pedestrian.

For the scenario where the car yields the way, there are three stages of the car's speed profile. First, the car reduces the speed (Fig. 6a), in order to provide a roomy distance to the pedestrian to cross. Second, the control model adjusts the speed in order to enable the car to leave as quickly as possible later while respecting a safety distance (Fig. 6b). When the pedestrian totally releases the crossing zone, the car quickly accelerates to reach the maximum speed limit. During the two first stages, the AV is able to come to a complete stop smoothly before the crosswalk, as priority is given to the pedestrian.

## V. CONCLUSION

In the present paper, we propose a new solution aiming at improving the safety and efficiency of pedestrian crossings involving autonomous vehicles. For this purpose, we rely on the communication abilities of the AV using both a light signal and an incentive speed profile. The control policy of the light signal and the vehicle speed profile is implemented using an innovative hybrid deep reinforcement learning model. According to the simulations, the proposed model offers the following benefits: fast convergence, the ability to separate decision and application in the reward, smoothness of the acceleration, and reactivity of the model over the pedestrian state.

These compelling results warrant further exploration through additional research efforts. Firstly, enhancing the smoothness of the curve generated by PPO and considering sparse rewards are of utmost importance. Secondly, extending the study to conflicts involving 2D crossings of pedestrian groups from multiple road lanes. Effectively tackling this issue necessitates the development of innovative signalling approaches, like displaying the remaining time and indicating the dangerous zone.

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